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BEYOND NEURAL

A QUANTUM APPROACH TO ARTIFICIAL COGNITION

DIPLOMA THESIS

BY

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Arta, October 2026

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Bibliography

Acknowledgements

I would like to express my gratitude to my supervisor, Mr Petros Karvelis, for his guidance and support during the preparation of this thesis.

I would like to extend special thanks to Mr Panagiotis Dimitrakis from the National Centre for Scientific Research 'Demokritos', as well as to Mr Georgios Kourousias from Elettra Sincrotrone Trieste, for the time, interest and scientific discussions they shared with me. Communicating with them was a valuable experience and contributed significantly to broadening my research perspective.

Finally, I would like to warmly thank my family, who supported me in completing this endeavour during my studies.

Abstract

This thesis explores the integration of cognitively inspired mechanisms into quantum computing, with an emphasis on quantum circuit optimisation. An architecture is proposed that combines algebraic circuit transformations with episodic memory, allowing for the recognition and recall of recurring structural patterns during the optimisation process.

The proposed method was evaluated on a structured parametric VQE circuit and a random circuit and compared with Qiskit's optimisation level 3. In the VQE circuit, a reduction of **17.20% in gate count** and **12.86% in depth** was achieved compared to Qiskit, with no observed change in Hellinger fidelity under the examined noise model. In the random circuit, the gate reduction was limited to **3.33%**, while the depth increased by **9.09%**, highlighting the dependence of effectiveness on circuit structure.

At the same time, a 500-iteration ablation experiment showed that episodic memory does not alter the final circuit's structural metrics and incurs a small time cost. Therefore, in this implementation, its function relates primarily to the recall and traceability of previous patterns rather than direct acceleration or the creation of structural compression.

The results present an early computational framework for investigating the relationship between selected cognitive mechanisms and optimisation transformations, without attributing full artificial cognition to the proposed architecture.

Keywords: Quantum Computing, Artificial Cognition, Quantum Circuit Optimisation, Episodic Memory, VQE, Qiskit, Transformations, Fidelity

1.1 Historical Overview and Definition of Artificial Intelligence

Artificial intelligence is not just one of the main subjects of interest for today's scientific community, but is the branch of computer science that studies the construction of computational systems with the ultimate expectation of mimicking the cognitive behaviour of human logic and logical reasoning. This concept is not limited to a specific technique or architecture. It examines different approaches to perception, reasoning, learning, decision making and problem solving. Russell and Norvig organise the different approaches to Artificial Intelligence around systems that think and act rationally, highlighting that there is no single way to approach the concept of intelligence in a computational system (Russell & Norvig, 2021).

One of the most important theoretical starting points for the study of artificial intelligence function appears in 1950 with the work of Alan Turing. Rather than attempting to provide an absolute definition of thought, Turing proposed studying the issue through an observable experiment, the imitation game. In this experiment, it is examined whether a human evaluator can, through written communication, distinguish a machine's behaviour from that of a human. In this way, the problem shifted from the philosophical definition of the concept of thought towards the evaluation of a computational system's behaviour (Turing, 1950).

The establishment of the term Artificial Intelligence is linked to the research proposal by John McCarthy, Marvin Minsky, Nathaniel Rochester and Claude Shannon in 1955 for the summer research meeting held at Dartmouth College in 1956. The basic hypothesis of the proposal was that features of learning and intelligence could be described with enough precision to make it possible for a machine to simulate them. The researchers set as subjects of study the use of language by computers, concept formation, problem solving and the possibility of machine self-improvement (McCarthy et al., 1955).

The transition from symbolic processing towards data-driven learning methods does not mean that the history of Artificial Intelligence is a simple linear replacement of one approach by another. On the contrary, the field has developed through different and often parallel research directions. Modern neural systems are a particularly important category of computational models, but the concept of Artificial Intelligence is not exclusively identified with them (Russell & Norvig, 2021).

This distinction is fundamental to this thesis. If Artificial Intelligence is treated as the broader research problem of building systems that exhibit complex cognitive and adaptive functions, then neural networks are a computational approach towards this goal and not the goal itself. Consequently, the investigation of different computational paradigms for information representation and processing remains scientifically open. This observation will form the basis for the subsequent examination of alternative computational approaches to artificial cognition.

1.2 From Symbolic Artificial Intelligence to Neural Networks

The evolution of Artificial Intelligence is characterised by different methods for developing techniques for information representation and processing. One of the most important historical distinctions concerns the symbolic approach and the connectionist approach, which is based on networks of simple computational units. The two approaches differ mainly in how information is represented and how computational processing is performed (Garson, 2019).

In the symbolic approach, information is represented through symbols and processing is performed through rules applied to specific representations. A system can, for example, possess a set of facts and rules and produce new conclusions by applying defined logical relationships. This approach was closely linked to the development of problem-solving systems, knowledge representation and expert systems. During the development of early Artificial Intelligence, symbolic processing was one of the dominant approaches for modelling complex cognitive functions (Russell & Norvig, 2021).

However, the aforementioned approach was not the only direction in the history of Artificial Intelligence. As early as 1943, Warren McCulloch and Walter Pitts presented a mathematical model of simplified neural units and investigated the possibility of forming computational networks from them. Their work formed one of the early theoretical bases for the development of artificial neural networks (McCulloch & Pitts, 1943).

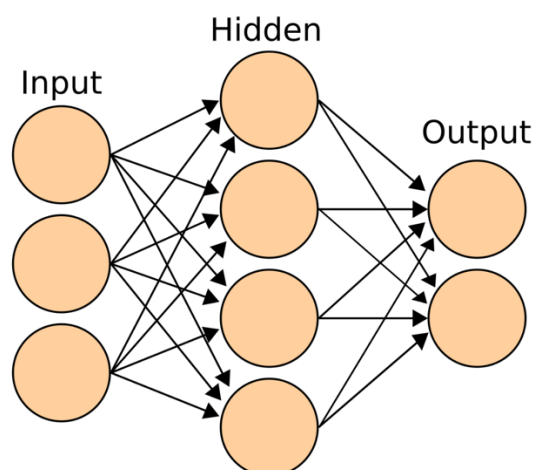


Figure 1.1 Basic structure of an artificial neural network

A later significant development was the perceptron model presented by Frank Rosenblatt. The model used numerical weights on its inputs and could adjust these parameters during training. This approach differs from a system in which behaviour is determined exclusively by a set of explicitly stated symbolic rules (Rosenblatt, 1958).

Interest in connectionist approaches was particularly strengthened during the 1980s. The work of Rumelhart, Hinton and Williams showed how error backpropagation to previous layers of a network can be used to adjust the internal representations of a multi-layer system (Rumelhart et al., 1986).

The connectionist approach should not be considered merely as a chronological successor to symbolic Artificial Intelligence. It is a different computational way of representing and processing information. In the symbolic approach, discrete representations and rules for manipulating them dominate, while in neural models, processing emerges from the dynamic operation of a network of connected units and from the values of the weights between them (Garson, 2019).

This distinction is particularly important for this thesis. The history of Artificial Intelligence shows that the computational representation of intelligence is not limited to a single paradigm. The transition from systems based primarily on explicit rules to systems that can adjust their internal parameters through learning is a prime example of a change in the computational framework. This observation creates the theoretical background for the investigation of different computational paradigms, which will be examined in the next chapters of this thesis.

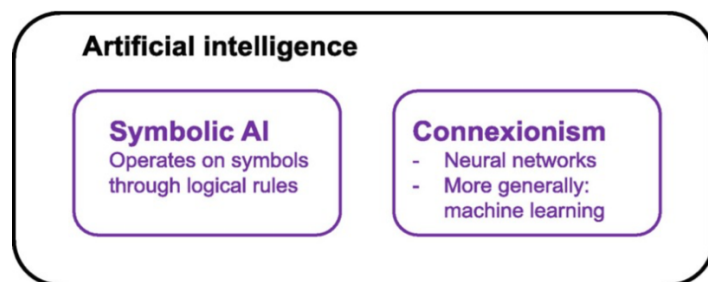


Figure 1.2 The two main families of approaches in Artificial Intelligence, the symbolic approach and connexionism.

1.3 Mathematical Structure of Artificial Neural Networks

Artificial neural networks consist of interconnected computational neurons organised in layers. Each neuron receives numerical inputs, after which appropriately designed mathematical transformations in the form of multidimensional vectors are applied to these inputs to produce the output. In feedforward networks, information is transmitted from the input to successive intermediate layers and finally to the output layer, without outputs returning to previous points of the same calculation (Goodfellow et al., 2016).

Mathematical representation of an artificial neuron

Let an artificial neuron can receive a set of inputs

$$x_1, x_2, \dots, x_n$$

Each input x_i is associated with a corresponding weight w_i . The neuron initially calculates the weighted sum of the inputs

$$z = \sum_{i=1}^n w_i x_i + b$$

Where

x_i is the value of the input with index i

w_i is the weight corresponding to the input x_i

n is the total number of neuron inputs

b is the neuron bias

z is the value resulting before the application of the activation function

The weight w_i determines how much the corresponding input influences the value calculated by the neuron. The bias b is an additional parameter of the affine transformation and allows for the shifting of the value produced by the weighted sum of the inputs (Goodfellow et al., 2016).

This specific operation in itself constitutes an affine transformation of the inputs. To construct neural networks capable of representing non-linear relationships, an activation function is subsequently applied (Goodfellow et al., 2016).

The final output of the neuron can therefore be written as

$$a = f(z)$$

where

f is the activation function

z is the weighted sum together with the bias

a is the output of the neuron after the application of the activation function

The activation function is therefore the stage that transforms the value z into the output a.

Activation functions

A classic activation function is the logistic function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where

z is the input of the function

e is the base of the natural logarithm

$\sigma(z)$ is the output of the function and takes values between 0 and 1

Another widely used function is the rectified linear unit, which is defined as

$$f(z) = \max(0, z)$$

If $z < 0$, the output is zero. If $z > 0$, the output equals z. Goodfellow et al. (2016) present rectified linear units as an important choice for the hidden layers of feedforward networks.

From a neuron to a layer

The operation of an entire layer can be described with vectors or, as we mentioned previously, transformations, for example:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

and the weights of a layer in a matrix $\mathbf{W}$ respectively, then the operation of the layer can be written as $\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}$ and then $\mathbf{a} = \mathbf{f}(\mathbf{z})$.

Multi-layer neural networks

A neural network can consist of successive layers. The output $\mathbf{a}$ of one layer is used as the input for the next. For a neural network, it holds that The exponent (l) denotes that a quantity belongs to layer l . Therefore $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ contain the parameters of the l -th transformation and the next layer the parameters of the next one respectively.

In general, it holds that: $\mathbf{a}^{(l)} = f^{(l)}(\mathbf{W}^{(l)} \mathbf{a}^{(l-1)} + \mathbf{b}^{(l)})$

For each layer of the neural network.

The weight matrices and bias vectors constitute the model's parameters. During training, the values of these parameters are adjusted with the aim of improving the network's performance with respect to a defined cost function. The process by which the error is calculated and the parameters are changed is the subject of the next section.

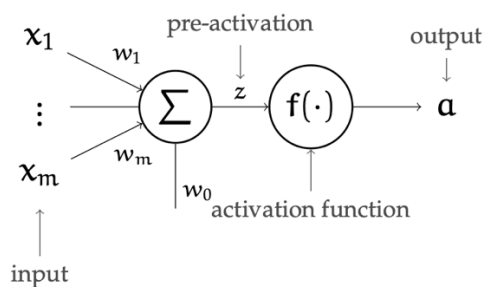


Figure 1.3 Mathematical structure of a basic artificial neural network computational unit, https://introml.mit.edu/_static/spring24/LectureNotes/chapter_Neural_Networks.pdf

1.4 Neural Network Training and Optimisation

The training of an artificial neuron is a process of continuous readjustment of the weights we discussed previously with the aim of reducing error, which in itself constitutes an optimisation problem in which the values of the parameters that minimise an appropriate cost function are sought (Goodfellow et al., 2016).

Let the set of trainable parameters of the network be denoted by $\boldsymbol{\theta}$ and the cost function by $\mathbf{J}(\boldsymbol{\theta})$, $\boldsymbol{\theta}$ includes the neuron's parameters such as weights and biases, while the quantity $\mathbf{J}$ provides the numerical measure of the model's performance

according to the criterion selected for the specific problem. Therefore, training is essentially the search for the appropriate parameter values to minimise the cost function

$$\theta^* = \operatorname{argmin} J(\theta)$$

Where θ^* denotes the set of parameters for which the minimum value of the examined cost function is achieved.

Gradient descent

A fundamental numerical optimisation method used in neural network training is gradient descent. The parameters are changed iteratively using information from the gradient of the cost function. The basic update rule can be expressed as

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} J(\theta_t)$$

where

θ_t are the parameters in the current iteration

θ_{t+1} are the parameters after the update

η is the learning rate

J is the cost function

$\nabla_{\theta} J$ is the gradient of the cost function with respect to the parameters θ is the iteration index.

The gradient is a vector with the partial derivatives of the cost function with respect to the model's parameters. It indicates the direction of the greatest local increase of the function. For this reason, the update is performed in the opposite direction when the goal is to reduce its value. The learning rate η determines the step size by which the parameters are changed in each iteration (Goodfellow et al., 2016).

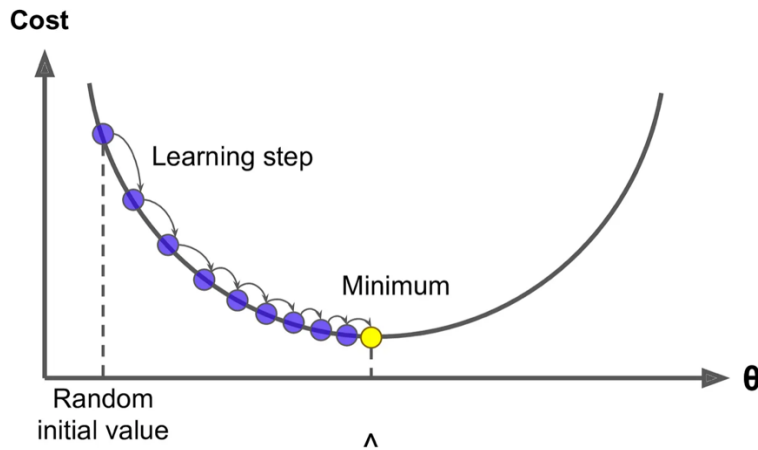


Figure 1.4 Graphical representation of the iterative change of parameters during gradient descent, <https://medium.com/@ckekula/batch-gradient-descent-in-machine-learning-a-mathematical-guide-7eeb6cac5e14>

The challenge of plateaus is worth mentioning, specifically the phenomenon of vanishing gradient, where the derivative of the cost function with respect to the parameters vanishes exponentially with the increase in the size of the system, making the use of specialised or adaptive optimisation architectures necessary.

Backpropagation of error

To perform the parameter update, the effect of each parameter on the final value of the cost function must be calculated. In multi-layer neural networks, this calculation is performed efficiently via backpropagation of error. The method utilises the chain rule of calculus so that the required derivatives are calculated from the network's output towards the previous layers (Rumelhart et al., 1986).

For a simple sequence of calculations, in which a weight w affects an intermediate value z , which in turn affects the activation a and finally the cost function J , the chain rule gives

$$\frac{\partial J}{\partial w} = \frac{\partial J}{\partial a} \frac{\partial a}{\partial z} \frac{\partial z}{\partial w}$$

The quantity $\frac{\partial z}{\partial w}$ expresses the effect of the weight w on the intermediate value z , the quantity $\frac{\partial a}{\partial z}$ the effect of z on the activation and the quantity $\frac{\partial J}{\partial a}$ the effect of the activation on the final value of the cost function. Their product allows the calculation of the total effect of the specific weight on the error.

The training of a neural network can therefore be summarised as an iterative process. Initially, the inputs are propagated forward through the network and an output is produced. Then the cost function is calculated. Through backpropagation, the derivatives with respect to the trainable parameters are calculated and, finally, the parameters are updated via the selected optimisation method.

The above process highlights a key characteristic of the neural approach. The system's behaviour depends to a significant extent on the numerical adjustment of its parameters based on the training data and the selected optimisation criterion. The computational execution of these operations is, however, carried out on specific computer architectures. The relationship between the model and the underlying computational architecture is examined in the following sections.

1.5 The Von Neumann Architecture and Memory Bottleneck

The operation of Artificial Intelligence algorithms depends not only on their mathematical formulation but also on the computational architecture on which they are executed. One of the most important models that influenced the development of modern computers is the Von Neumann architecture, which is associated with the stored-program principle. Key elements of this principle were described in the first design report of the EDVAC electronic computer in 1945, while the historical development of the idea also included significant contributions from J. Presper Eckert, John Mauchly and Arthur Burks (IEEE Computer Society, n.d.).

In a simplified representation, such a computational system consists of the central processing unit, the memory and the input and output mechanisms. The central processing unit performs arithmetic and logical operations and coordinates the execution of instructions, while the memory stores both data and instructions required during the execution of a program.

The distinction between processing and storage is of particular importance in applications that require the processing of large quantities of data. The performance of a computational system can be limited not only by the maximum computational capacity of the processor but also by the rate of data transfer between memory and the processing unit. This problem is associated with the so-called Von Neumann bottleneck (Backus, 1978).

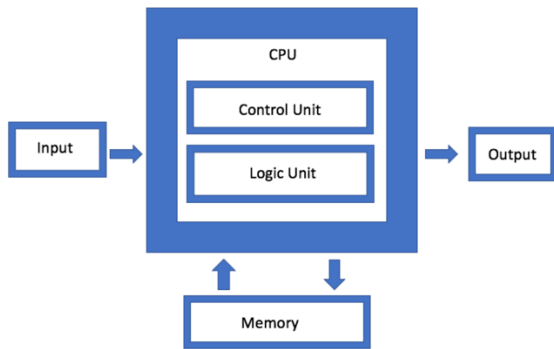


Figure 1.5: The Von Neumann architecture

The relationship between computational performance, memory bandwidth and arithmetic intensity can be described via the roofline model. The achievable performance of a system can be expressed as

$$P = \min(P_{\max}, B \cdot I)$$

With P as achievable computational performance, $P_{\max}$ maximum computational performance of the system, B the bandwidth and I the arithmetic intensity, i.e., the volume of operations applied to the data and transferred to the memory. This relationship shows that increasing the processor's power does not necessarily imply a corresponding increase in actual performance. When the product $B \cdot I$ is smaller than $P_{\max}$, the performance is limited mainly by data transfer. When it is larger, the limitation shifts to the computational capacity of the processor (Williams et al., 2009).

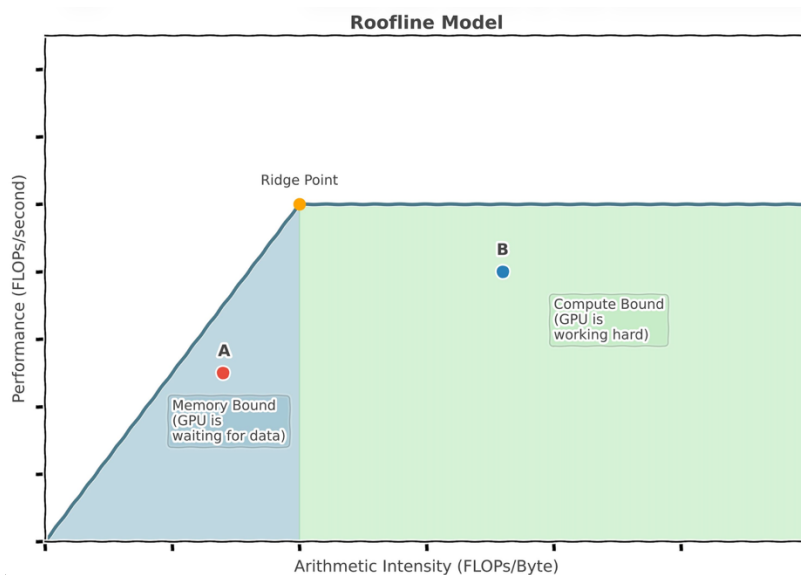


Figure 1.6 Schematic representation of the relationship between arithmetic intensity, memory bandwidth and maximum computational performance.

This issue is particularly relevant to neural networks, as their training and execution involve repetitive transfer of parameters, activations and intermediate results. The energy and time overhead associated with data movement is therefore a significant parameter in the design of modern computational systems for Artificial Intelligence applications (Sze et al., 2017).

1.6 Computational Constraints and the Need for Alternative Computational Paradigms

The increase in the scale of modern computational models has strengthened interest in architectures that are not limited exclusively to the established way of organising memory and processing. Research in computing has led to the development of different paradigms, such as neuromorphic and quantum computing, which differ in terms of the way information is represented and processed (Marković et al., 2020).

Quantum computing, which will be analysed more extensively in a later chapter, is based on quantum states and transformations described by the laws of quantum mechanics. It is not simply a faster version of classical computing and does not imply a general acceleration of every computational problem. Instead, it is a different model of computation, in which specific classes of problems can exhibit different computational properties in relation to classical approaches (Nielsen & Chuang, 2010).

In conclusion, the investigation of alternative computational paradigms therefore becomes interesting not as a direct replacement for neural networks, but as a search for different ways of representing, transforming and utilising information. Within the framework of this work, this direction creates the theoretical basis for the study of artificial cognitive functions that are not limited exclusively to the neural computational paradigm.

Chapter 2: Artificial Cognitive Function and Alternative Computational Paradigms

2.1 The Concept of Artificial Cognitive Function

When we refer to the term Cognitive function, we do not refer exclusively to the methodology of drawing conclusions of the human brain, but also to a set of processes related to the reception, processing, storage and retrieval of information with the aim of its utilisation by a computational system. These include several of them, among which are perception,

attention, memory, learning, reasoning as well as the selection of final action. So-called Cognitive science deals precisely with these functions as well as with their interactions in information processing.

Computational study cannot guarantee the complete imitation of these functions by a computational system or that it possesses consciousness. On the contrary, it examines how they can be described through computational mechanisms and representations, with Cognitive architectures being a classic example of one of these approaches; this is because they attempt to integrate functions as we mentioned within a unified computational framework. The review by Kotseruba and Tsotsos records these abilities as key characteristics used for the comparison of different cognitive architectures.

Within the framework of this work, the term **artificial cognitive function** is used in a functional sense. It does not refer to the complete reproduction of human intelligence, but to the ability of a computational system to maintain information regarding its state, to utilise previous results and to adjust its subsequent actions according to the conditions it faces. This specific approach allows the study of cognitive characteristics without requiring the adoption of a specific biological or neural architecture.

We consider this distinction important for the present work, as the research interest which we will analyse subsequently is not limited to the creation of a model that produces a desired output but also to the way it maintains an internal state, takes into account previous changes and utilises them in order to modify its behaviour through feedback. In this sense, artificial cognitive function is treated as a problem of organisation and utilisation of information and not exclusively as a problem of training a neural model.

This specific perspective creates the theoretical background for the investigation of different computational paradigms. Since cognitive functions such as memory, action selection and adaptation can be studied at the level of computational architecture, the question arises as to whether they can also be implemented through non-neural mechanisms. The investigation of this possibility constitutes one of the key axes of the following sections.

2.2 Intelligence and Cognitive Function

Intelligence and cognitive function are closely linked concepts, without, however, being identical. The study of intelligence has led to different theoretical approaches and ways of measurement. In the psychometric approach, the concept of general intelligence holds a special place, which expresses the common variance observed in performance between different mental tests (Barbey, 2021).

Cognitive function can be examined at the level of individual abilities and mechanisms. In the study of cognitive architectures, functions such as perception, attention, action selection, memory, learning, reasoning and

metareasoning are encountered. These functions do not necessarily appear in the same way in every architecture, but are used as key axes for the study and comparison of cognitive function computational systems (Kotseruba & Tsotsos, 2020).

The relationship between individual cognitive functions and intelligence is of particular interest. Reasoning, for example, is closely related to intelligence, without the two concepts being considered fully identical. Correspondingly, working memory allows for the temporary maintenance of information relevant to a goal during other cognitive processes and is linked to functions such as reasoning, planning and problem solving. Its capacity also shows a strong correlation with general fluid intelligence (Conway et al., 2020).

From a computational point of view, this distinction is important. A system can exhibit high performance in a specific task without this implying that it possesses a broader set of cognitive functions. On the contrary, the study of a cognitive architecture examines the way in which different functions can coexist and interact within the same system. The literature on cognitive architectures actually shows that there is no single accepted architecture for the representation of these functions and that different approaches use different mechanisms of representation, memory, learning and reasoning (Kotseruba & Tsotsos, 2020).

Within the framework of this work, intelligence is not treated as an isolated computational function. The interest is focused on specific cognitive abilities and mainly on the way in which an artificial system can maintain information, learn from previous states and use this information during decision making. These three functions are the subject of the next section.

2.3 Memory, Learning and Decision Making

Memory as a key component of cognitive systems allows for the maintenance of information; what is done so effortlessly and naturally by the human brain is not used exclusively as a data storage space in a computational system, but also in the maintenance of intermediate results and in the adjustment of its behaviour. For this reason, different forms of memory appear in the majority of cognitive architectures that have been proposed in the literature (Kotseruba & Tsotsos, 2020).

A key distinction concerns the duration and the type of information maintained. In the literature on cognitive architectures, forms of short-term and long-term memory are encountered, as well as distinctions between declarative, procedural and semantic memory. These categories are not necessarily implemented as fully independent subsystems, as their precise organisation depends on the design and goals of each architecture (Kotseruba & Tsotsos, 2020).

The importance of **memory** becomes more apparent when examined in combination with learning. Memory allows for the maintenance of previous information, while **learning** allows for the change of existing knowledge or behaviour based on new experiences. The two functions are therefore closely linked, as without the maintenance of previous information, each new situation would be dealt with independently of the previous ones. Correspondingly, without a learning mechanism, the stored information could not by itself lead to adaptation to new conditions (Kotseruba & Tsotsos, 2025).

The **decision making** complements this relationship, as it concerns the selection of an action among available alternatives based on the current state and the information available to the system. In cognitive architectures, action selection is recognised as one of the key cognitive abilities along with memory, learning, perception and reasoning (Kotseruba & Tsotsos, 2020).

A characteristic example is the **ACT-R** cognitive architecture. In it, different functions are organised into individual modules, among which are modules of declarative memory, goals, and perception and action. Information from the modules becomes available through temporary structures and a system of production rules selects the appropriate action depending on the current state. The architecture therefore shows how stored information, internal state and action selection can constitute interdependent parts of a unified cognitive system (Anderson et al., 2004).

For the present work, this interaction is of particular importance. A system that maintains information about previous states can use this information during the evaluation of a new situation. The result of an action can subsequently change the system's internal information and influence its subsequent choices. A general cycle thus emerges

κατάσταση → αξιοποίηση μνήμης → επιλογή δράσης → αποτέλεσμα → ενημέρωση

The above relationship does not in itself constitute a specific model of human cognition; it is a functional description of the way in which the maintenance of information can be linked to selection and adaptation. This distinction is important for the present work, as it allows for the examination of such properties independently of the requirement of their implementation through an artificial neural network.

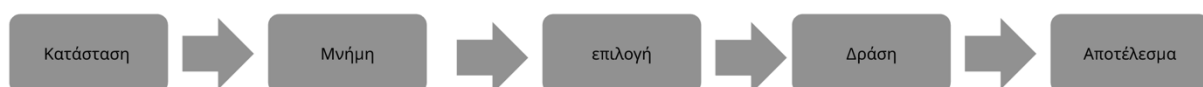


Figure 2.1 Simplified representation of the cycle of memory, action selection and internal state update.

2.4 Internal State and Dynamic Evolution

The ability to maintain an internal state constitutes an important characteristic of a system that is called upon to retrieve information from its memory during decision making. The behaviour of such a system is not determined exclusively by random information it received at a specific moment in time but by its perception of previous states as well as what its observations and actions were. In cognitive architectures, the maintenance and update of internal representations is linked to functions such as working memory, learning and action selection (Kotseruba & Tsotsos, 2020).

The general mathematical representation is denoted by the column vector

$$\mathbf{s}_t = \begin{bmatrix} s_{1,t} \\ s_{2,t} \\ \vdots \\ s_{n,t} \end{bmatrix}$$

where $\mathbf{s}_t$ is the total internal state of the system, $s_{i,t}$ is the value of the i -th state variable at time t and n is the total number of variables used for the description of the state. This representation is general and does not presuppose that the variables correspond to artificial neurons.

Naturally, as the system interacts with its environment, its internal state is also readjusted. This evolution can be described by the transition function

$$\boxed{\mathbf{s}_{t+1} = F(\mathbf{s}_t, \mathbf{x}_t, \mathbf{a}_t)}$$

where $\mathbf{s}_{t+1}$ is the next state, $\mathbf{s}_t$ the previous state, $\mathbf{x}_t$ the information received at time t , $\mathbf{a}_t$ the action performed and F the transition function towards the new internal state.

The above relationship expresses an important property. The next state does not necessarily depend only on the current input, but also on the state in which the system was already in, therefore the same input information can lead to a different evolution when the system has a different previous history.

A similar logic to the above is found in partially observable Markov decision processes. When the actual state of the environment is not directly known, a belief state can be used, which represents a probability distribution over the possible states of the environment. The state

is updated as new observations and actions become available (Kaelbling et al., 1998).

The choice of an action can then be described through a policy which can vary depending on the system and its design

$$\mathbf{a}_t = \pi(\mathbf{s}_t)$$

Where π is the action selection rule, $\mathbf{a}_t$ the action selected based on the current state. After taking the action, the function F updates the state and the loop repeats.

When the consequences of a decision extend over more time steps, the total discounted reward can be used

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$$

where G_t is the total future reward from time t , r_{t+k+1} is the reward received after $k+1$ time steps and γ is the discount factor with

$$0 \leq \gamma \leq 1.$$

The parameter γ determines the relative importance of future rewards. Smaller values place greater emphasis on immediate consequences, while values closer to unity increase the weight of later results (Sutton & Barto, 2018).

This specific description does not constitute a model of human cognition, but a general computational formulation of a system with an internal state. For this work, it is particularly important, as it shows that properties such as the retention of previous information, action selection, and adaptation can be described without assuming the use of a neural network. Thus, the basis is created for examining different computational mechanisms for the representation and dynamic evolution of artificial cognitive states.

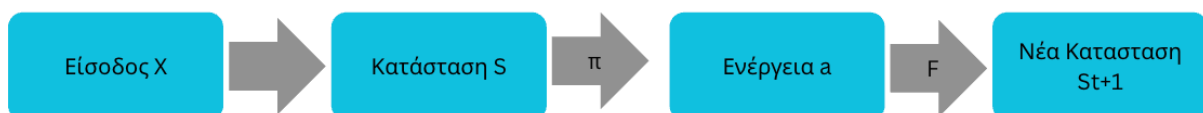


Figure 2.2 Simplified representation of the dynamic evolution of the internal state and action selection.

2.5 Uncertainty and Adaptability

A cognitive system does not always have complete information about the environment in which it is placed into operation. Additionally, observations may not contain clear information; noise and omissions are added to the observations, and the result of an action may not be known in advance. For this reason, decision-making under uncertainty is an important part of the study of cognitive function.

The mathematical representation of the uncertainty we refer to can be captured through a probability distribution which naturally obeys the probability normalization rule:

$$P(s_i) \geq 0 \text{ and } \sum_{i=1}^n P(s_i) = 1.$$

For every possible state in the set $S = \{s_1, s_2, \dots, s_n\}$

S expresses the set of possible states, the s_i a specific state and $P(s_i)$ the probability assigned to it. In this way, the system does not need to consider only one state as certain, but can maintain different probabilities for multiple possible states.

Similar approaches are used in Markov decision processes, where when a state for a specific environment is not certain, we use a **belief** state, which is part of the set of probability distributions over all possible states of the system. As new observations become available, this distribution can be updated and used for selecting subsequent actions (Kaelbling et al., 1998).

where b_t is the belief state at time t ,

a_t the action that was performed, O_{t+1} the new observation and τ the update process that produces the new belief state. b_{t+1}

Adaptability is a method that allows the system itself, when it receives new information, to be able to adjust its future state according to the data it receives from it and to change the estimate it has from already existing data of the current environment state. Thus, its behaviour is adapted according to the information acquired during interaction with the environment.

Regarding Artificial cognitive function, this capability is considered particularly important so that the system that possesses memory and an internal state can revise its internal representation and alter its subsequent decisions.

Managing uncertainty is therefore one of the mechanisms through which adaptive behaviour can emerge.

2.6 Limitations of Neural Approaches and Alternative Computational Paradigms

Artificial Neural Networks are one of the main pillars of engagement and active research in the field of Modern Artificial Intelligence and have shown high performance in both recognition problems and data prediction and processing problems. However, high performance in a specific task does not necessarily imply the existence of more general cognitive abilities. Issues such as generalisation to new situations and systematic reasoning remain significant research problems (Lake et al., 2017).

An additional limitation concerns interpretability. In complex neural models, it is not always easy to determine why the system produced a specific decision. This issue has led to the development of interpretable Artificial Intelligence methods, with the aim of better understanding the behaviour of models (Arrieta et al., 2020).

However, increasing the size of the hyperparameters of a quantum model, as we discussed in the previous chapter, can be accompanied by a simultaneous and exponential increase in increased resource, memory, and data transfer requirements from memory to computational units; such factors significantly affect both the execution speed and the energy consumption of the system (Sze et al., 2017).

These limitations do not negate the usefulness of neural networks. They do, however, show that artificial cognitive function can also be examined through different computational paradigms. Symbolic Artificial Intelligence, for example, uses explicit knowledge representations and reasoning rules, while other approaches use different mechanisms for representation and information processing.

Of particular interest for this work is **quantum computing**, as it uses a different mathematical and physical formalism for the representation and transformation of information. The state of a quantum system, superposition, entanglement, and measurement create representation possibilities that differ from corresponding classical approaches (Nielsen & Chuang, 2010).

The existence of these different fields shows that functions such as memory, adaptation, uncertainty management, and decision-making do not need to be examined exclusively through neural architectures. In the context of this work, the interest therefore shifts to investigating whether a different computational model, based on quantum principles, can be used for the representation and evolution of artificial cognitive states.

2.7 Quantum Models of Cognitive Processes and Approaches to Artificial Cognitive Function

Quantum Models of Cognitive Processes use the mathematical and physical properties of quantum mechanics in combination with probability theory to describe phenomena such as uncertainty, the dependence of a decision on context, and the effect of previous judgements on subsequent ones. This approach, of course, does not assume that the human brain is identical to a quantum system but that it uses this formalism as a different way of representing certain cognitive states (Pothos & Busemeyer, 2022).

Such a cognitive state is connected via a multidimensional column vector in Hilbert space as

$$|\psi\rangle = \sum_{i=1}^n c_i |s_i\rangle$$

Where the ket state is the current system state with a corresponding ket vector $|s_i\rangle$ of the possible states and the probability amplitudes which follow normalization

$$\sum_{i=1}^n |c_i|^2 = 1.$$

The probability of a specific result occurring can be calculated according to the Born rule

$$P(s_i) = |\langle s_i | \psi \rangle|^2.$$

Where $\langle s_i | \psi \rangle$ is the probability amplitude and the square expresses its modulus, i.e., the projection onto the state of the state.

This specific representation differs from a simple classical probability distribution. Amplitudes can include phase information, with the result that different possibilities can exhibit **interference**. At the same time, different cognitive questions or measurements can be represented by non-commutative operators. In the case of two operators A and B, it holds that i.e., it means that the order of their application can affect the result. This property has been used in quantum cognitive modelling for phenomena such as the effect of question order, decision-making under uncertainty, and context dependence (Pothos & Busemeyer, 2022).

Characteristic	Classical approach	Quantum approach
State	Variables or probabilities	State vector
Uncertainty	Probability distribution	Probability amplitudes
Update	Classical probabilistic update	Transformation and measurement
Interference	Not a general characteristic of classical probability	Can occur between amplitudes
Order of decisions	Not inherently order-dependent	Can depend on the order of non-commutative operations
Context	Incorporated into the model with additional variables or dependencies	Can be incorporated into the geometry of states and measurements

Table 2.1 Comparison of classical and quantum representation *based on Pothos and Busemeyer, 2022 and Khrennikov, 2015.*

However, it is important to note once again that several quantum-inspired cognitive models can describe existing experimental results, but their actual cognitive predictive capability remains an open problem for investigation.

For this work, therefore, the interest lies in a specific question. Can properties such as **state, memory of previous interactions, uncertainty, selection, and adaptation** be organised into a non-neural architecture that uses quantum representation and information evolution? This question forms the connection between the theoretical background of this chapter and the architecture that will be investigated experimentally next.

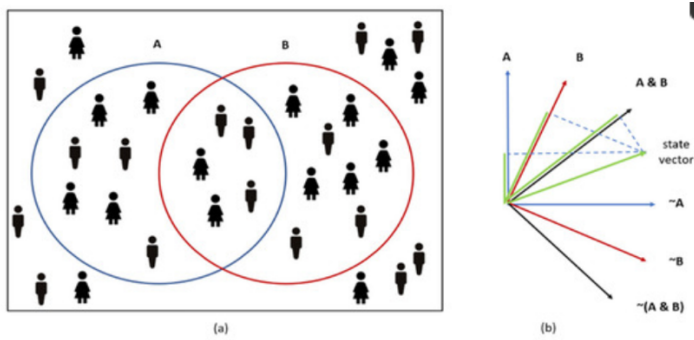


Figure 2.3 Geometric representation of a quantum-type state and its possible projections in a space of states in combination with different set operations.<https://www.mdpi.com/2504-2289/6/4/135>

3.1 Introduction to Quantum Computing

Quantum Computing is not just a different model of information processing, but a completely different perspective on computation described by the fundamental rules of quantum mechanics. Unlike classical computers where the basic unit of information, the Bit, is represented through deterministic binary digits. Their operation leverages properties such as quantum superposition, entanglement, and interference, which allow for different ways of representing and transforming information (Nielsen & Chuang, 2010).

The basic idea is that a quantum computer does not simply differ in terms of capacity or speed, nor is it a new type of system that is coming to replace the classical processors used today. It is a different computational model, in which information is encoded in quantum states and changed through controlled quantum operations. Measurement at the end of a calculation provides a classical result, and therefore a quantum algorithm must be designed so that the evolution of the quantum state increases the probability of extracting useful information.

Regarding Research interest, quantum information systems focus on specific classes of problems in which quantum algorithms can offer different or more efficient computational approaches. At the same time, today's quantum systems remain sensitive to noise and errors, a fact that limits the size and complexity of calculations that can be performed reliably.

For this work, of particular interest is not only the potential computational speed-up but mainly the **different way of representing and evolving information**. To understand this difference, it is necessary to first examine the basic unit of quantum information and the mathematical description of the quantum state.

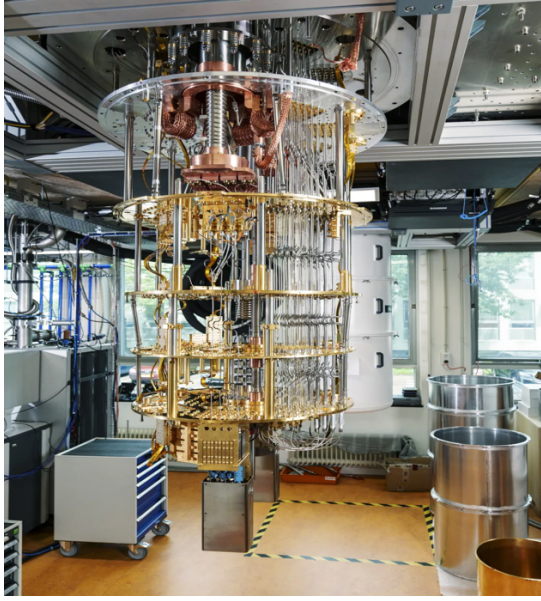


Figure 3.1 Internal layout of a dilution refrigerator used for the operation of superconducting quantum systems at extremely low temperatures. Source QuTech.

3.2 Quantum Information Unit, States and Superposition

The quantum binary digit, or Qubit, is the basic unit of information in quantum computing. In a classical system, a binary digit can be in one of two states, 0 or 1. Conversely, the state of a quantum binary digit can be described as a linear combination of the two basis states and $|0\rangle$ and $|1\rangle$ (Nielsen & Chuang, 2010).

The two basis states of a Qubit are

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

A general state of a quantum binary digit is described as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Where $|\psi\rangle$ is the total state and α, β are the complex probability amplitudes and not the probabilities themselves; to find the probabilities, we take the modulus of each amplitude as we saw in the previous chapter.

Also, for a normalised state, it holds

$$|\alpha|^2 + |\beta|^2 = 1.$$

during measurement in the computational basis, the results 0 and 1 respectively (Nielsen & Chuang, 2010).

Quantum superposition

When $\alpha, \beta \neq 0$ holds, then the Qubit is in **superposition** between the states $|0\rangle$ and $|1\rangle$. For the sake of the example, we have the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

In this case

$$P(0) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}, \quad P(1) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$

Therefore, a measurement in the computational basis gives 0 or 1 with equal probability. However, superposition should not be interpreted simply as classical ignorance regarding whether the system is in 0 or 1. The amplitudes α and β can possess a relative phase, which affects the future evolution of the system and allows for interference phenomena.

A state of a Qubit can also be written as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

The parameter θ determines the relative contribution of the two basis states, while ϕ describes the relative phase between them. The factor i is the imaginary unit, for which it holds. This formula allows for the geometric representation of a state of a quantum binary digit through the **Bloch sphere**.

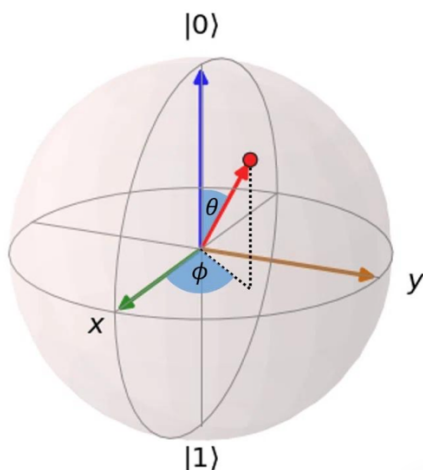


Figure 3.2 Geometric representation of the state of a quantum binary digit through the Bloch sphere.

On the Bloch sphere, the north pole corresponds to the state, the south to, while every point on the surface represents a pure state of a quantum binary digit. The angles θ and ϕ determine the position of the state on the sphere.

3.3 Quantum Measurement

As Quantum measurement, we define the process during which a qubit ceases to be in the process of superposition and gives an observable classical result.

That is, let a Qubit be in state then a measurement in the computational

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

basis can give either 1 or 0.

The probabilities of the two outcomes are given by the Born rule and are

$$P(0) = |\alpha|^2, \quad P(1) = |\beta|^2.$$

After measurement, if the result is 0, the system is in the state $|0\rangle$. Correspondingly, if the result is 1, it is in the state $|1\rangle$. Therefore, measurement is not merely a reading of an already known classical value, but is part of the mathematical description of the quantum system itself (Nielsen & Chuang, 2010).

Let us provide an example to better understand this concept mathematically

Let the state be

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle \quad \text{and measurement probabilities}$$

$$P(0) = \left| \frac{\sqrt{3}}{2} \right|^2 = \frac{3}{4}, \quad P(1) = \left| \frac{1}{2} \right|^2 = \frac{1}{4}.$$

Thus, a single measurement can return 0 or 1, but if the same state is prepared and measured many times, the results are expected to appear in a ratio of approximately 75% to 25%.

In a more general form, if the measurement is described by a set of projector P_m the probability of the result m can be written as

$$P(m) = \langle \psi | P_m | \psi \rangle$$

Where $|\psi\rangle$ is the state before measurement, P_m is the projector of the result m . This relationship generalises the Born rule beyond simple measurement in the $|0\rangle$ and $|1\rangle$ states.

Measurement has particular significance in quantum algorithms because it is the point at which quantum processing is converted into classical information. For this reason, the evolution of a quantum circuit must be designed so that the desired answers appear with the highest possible probability during measurement.

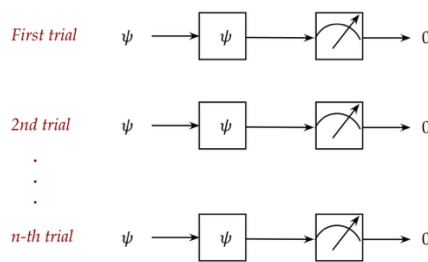


Figure 3.3 Schematic representation of quantum bit measurement in the computational basis, <https://www2.seas.gwu.edu/~simhawe/quantum/modules/module3/module3.html>

3.4 Quantum Entanglement and Interactions of Multiple Quantum Bits

After the superposition of a quantum bit, a reasonable question arises regarding its reaction with the remaining computational units (Qubits); this question relates to how multiple Qubits can react to a series of operations and transformations, and **Quantum Entanglement** comes to solve this query.

Suppose, therefore, that we have 2 qubits and available states where $n=2$, so we have available states

$$2^n$$

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle.$$

And general state

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

With $\alpha, \beta, \gamma, \delta$ as complex probability amplitudes, normalised.

Quantum entanglement occurs when the state of the total system cannot be written as a simple product of two independent states. A characteristic example is the Bell state.

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

which cannot be decomposed into the form of a tensor vector. $|\psi_A\rangle \otimes |\psi_B\rangle$
This means that the information describing the total system cannot be fully attributed to two independent subsystems. If a measurement is performed in the computational basis, the results 00 and 11 appear with a probability of 1/2 each, while 01 and 10 do not appear.

The state can be created initially by applying a Hadamard gate as a gate that creates superposition on the first Qubit and then a controlled-NOT gate between the two quantum bits

$$|00\rangle \xrightarrow{H \otimes I} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{\text{CNOT}} \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

The Hadamard gate activates the superposition on the second Qubit, and the identity vector **I**, as we already know from basic principles of linear algebra, does not perform any change; conversely, the controlled-NOT gate (CNOT) flips the second Qubit when the first is in the state $|1\rangle$, leaving us with only two possible states, $|00\rangle$ and $|11\rangle$.

Entanglement is an important resource in quantum information because it allows correlations that cannot be described as simple independent classical probabilities. At the same time, the creation of such correlated states requires interactions between quantum bits. In real quantum systems, these interactions are limited by the physical connectivity of the device and by the quality of the available two-quantum-bit gates. These limitations will become particularly significant in the following sections on the depth and compilation of quantum circuits.

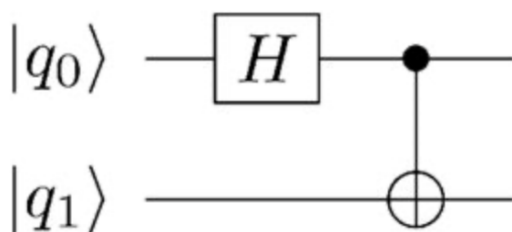


Figure 3.4 Creation of an entangled Bell state via a Hadamard gate and a controlled-NOT gate, <https://quantumcomputing.stackexchange.com/questions/16341/what-gate-combinations-create-entangled-two-qubit-states>

3.5 Quantum Gates and Circuits

Quantum gates, as mentioned in previous chapters, are mathematical linear transformations that mutate the state of a quantum system. Mathematically, they are described by unitary operators so that the normalisation of the quantum state is maintained. For a unitary operator U , the following holds

$$U^\dagger U = I$$

where $U^\dagger$ is the conjugate transpose of U and I is the identity matrix. If the initial state is $|\psi\rangle$, after applying the gate, we obtain

$$|\psi'\rangle = U|\psi\rangle.$$

the Hadamard gate is represented mathematically as the vector

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Where the mathematical formalism for representing its application is

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Two-quantum-bit gates are of particular importance because they allow the creation of interactions and entanglement between different parts of a quantum system. A characteristic example is the CNOT. The first input acts as a control and the second as a target. When the Qubit is in the state $|1\rangle$, the target state is flipped.

A quantum circuit is formed by a temporally organised sequence of such gates, measurements, and, when required, other operations on a set of quantum bits. The order of the gates is important, as different operations do not necessarily commute, and therefore different sequences can produce different final states.

In practice, however, a theoretical circuit cannot always be executed exactly as is on a real quantum device. The available hardware supports a specific set of basic gates and specific physical connectivity between the quantum bits. For this reason, compilation rewrites the original circuit so that it is compatible with the topology and available operations of the device.

Multi-quantum-bit gates are also of particular importance for circuit optimisation. IBM itself points out that multi-quantum gates exhibit higher error rates and that reducing the total number of gates is one of the goals of optimisation during compilation. For this reason, the number and position of two-quantum-bit interactions are important parameters when evaluating a quantum circuit.

A special category is **Parametrised Quantum Circuits**, which include variable gates of the form $U(\theta)$. The specific gate sequence structure defines an **ansatz**, i.e., a parametric form of a quantum state $|\psi(\theta)\rangle$, which is subject to optimisation via external algorithms.

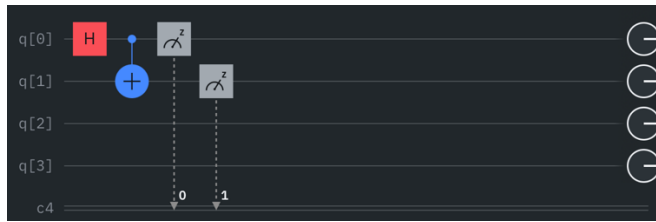


Figure 3.5 Example of a quantum circuit consisting of one Hadamard gate, one CNOT, and two-Qubit measurement gates using IBM's Quantum Composer.



Figure 3.6 Probability distribution for the above circuit using IBM's Quantum Composer.

3.6 Noise and Limitations of Modern Quantum Systems

The operation of a quantum system operating under real conditions differs from the ideal mathematical description of a quantum circuit. Qubits interact with their environment, applicable gates are not executed with absolute precision, and measurements can be noisy and have an error rate; this indicates that there is a gap between the mathematical representation and the computational application. For this reason, noise is one of the most significant limitations of current quantum systems (Preskill, 2018).

Particularly important is **decoherence**, during which the quantum state gradually loses the properties necessary for computation. The times and

are used to describe this behaviour. T_1 is related to energy loss, while T_2 is related to the loss of coherence of the phase of the quantum state (Kjaergaard et al., 2020).

Despite the active efforts that the research and industrial sector have made to date to mitigate error per quantum gate, we believe that this specific area of quantum computing can still receive significant assistance so that execution results are viable.

However, every real quantum gate presents a given error rate. We can define this error with a simplified mathematical model: with a gate where it has an error probability ϵ_i , then the probability of successful execution of a sequence of N gates can be mapped as follows:

$$P_{\text{επιτυχίας}} \approx \prod_{i=1}^N (1 - \epsilon_i)$$

Where N is the number of gates, and ϵ_i is the error of each gate, normalised probability, which of course does not constitute a complete model of real quantum noise. However, it shows a basic idea. **The more non-ideal operations a circuit requires, the more opportunities there are for error accumulation.**

Factor	What it causes
Decoherence	Gradual loss of the quantum state
Gate errors	Deviation from the desired operation
Measurement errors	Incorrect reading of the result
Limited connectivity	Need for additional operations between non-adjacent quantum bits
Large depth	Longer duration and exposure of the circuit to noise

Table 3.1 Basic sources of limitations in modern quantum systems. Own processing based on Preskill, 2018 and Kjaergaard et al., 2020.

Two-quantum-bit gates are of particular interest, as they are necessary for creating entanglement but are usually more demanding than single-quantum-bit gates on current platforms. The physical connectivity of the device also determines which quantum bits can interact directly. Thus, a theoretically correct circuit may in practice require additional gates to adapt to the real hardware (Kjaergaard et al., 2020).

These limitations explain why the quality of a quantum circuit does not depend only on the result it calculates. The number of gates, the execution duration, the

interactions, and the topology of the device also affect its practical reliability. For this reason, the **depth and complexity of the circuit** are important design and optimisation criteria.

To overcome these limitations, **Hybrid Quantum-Classical Algorithms** are used, such as VQE and QAOA. In this example, the quantum system undertakes the calculation of expected values, while a classical optimiser recursively updates the parameters θ , minimising the effect of noise.

For evaluation, however, we must also take into account the presence of noise, i.e., comparing the result distribution of a noisy and an ideal circuit; this evaluation is performed by Hellinger fidelity.

$$F_H(P, Q) = \left(\sum_x \sqrt{P(x) Q(x)} \right)^2$$

Where $P(x)$ and $Q(x)$ are the probabilities of the result x in an ideal and noisy state, respectively. We accept the expected value 1 (high similarity) as a result, while lower values indicate greater deviation between the distributions (Hellinger, 1909).

3.7 Depth and Complexity of Quantum Circuits

The complexity of a quantum circuit does not depend only on the number of quantum bits it uses. The number of gates and the way in which they are distributed during the execution of the circuit also play an important role. Two circuits can therefore perform the same calculation but differ significantly in terms of their execution cost.

A basic metric is the **circuit depth**. Depth expresses the number of successive layers of operations, taking into account that gates acting on different quantum bits can in some cases be executed in parallel.

For example

$$D(C) = \max_{q_i} d(q_i)$$

where C is the quantum circuit, q_i a quantum bit and $d(q_i)$ is the number of successive layers of operations that affect its path. $D(C)$ therefore expresses the depth of the total circuit.

A second useful metric is the **number of gates**

$$G(C) = \sum_{k=1}^m N_k$$

Where it describes the sum of occurrences of gate N_k and m is the number of different gates being examined.

The two metrics are not identical. A circuit can contain many gates but have a relatively small depth, provided that several of them can be executed in parallel. Conversely, a long sequence of dependent operations increases the depth even when the total number of gates is not particularly large.

Metric	What it describes	Why we care
Number of gates	Total count of operations	Computational cost and potential error accumulation
Depth	Successive layers of operations	Duration and exposure to noise
Two-quantum-bit gates	Interactions between pairs	Increased practical cost in many current architectures
Connectivity	Which pairs can interact directly	Affects additional operations during compilation

Table 3.2 Basic characteristics that affect the complexity of a quantum circuit.

Reducing only one of these metrics does not always guarantee a better circuit; for example, a transformation might reduce the depth but increase the number of certain gates. For this reason, quantum circuit optimisation can be treated as a problem with **more than one criterion**, depending on the limitations of the device on which the circuit is to be executed (Sivarajah et al., 2020).

This is precisely what creates the need for **compilation and optimisation of quantum circuits**, where an initial circuit is transformed with the goal of more efficient execution on real quantum hardware.

3.8 Compilation and Optimisation of Quantum Circuits

The design of a quantum circuit may not always follow the physical limitations and properties of the device on which it will be executed. In practice, however, every quantum processor has a specific topology, gates, and limitations regarding which Qubits can interact directly; that is, there is a clear separation of Qubit topologies and logical Qubits. For this reason, before execution, the process of **compilation** of the circuit is required (Sivarajah et al., 2020).

During compilation, the original circuit is converted into a form compatible with the specific quantum device. One of the basic problems is the mapping of logical quantum bits to the physical quantum bits of the processor. If two logical quantum bits must interact but are not in suitable physical positions, the introduction of additional routing operations, such as swap gates, may be required (Li et al., 2019).

A simple example is

$$q_0 \leftrightarrow q_1 \leftrightarrow q_2$$

If, therefore, interaction is required between q_0 - q_2 , but the device only allows connections q_0 - q_1 and q_1 - q_2 , the circuit must adjust the topology by adding extra gates to swap positions.

This process can increase the number of gates and the depth of the circuit. Therefore, after or in parallel with compilation, **optimisation** is performed, with the goal of producing an equivalent circuit with a lower practical cost.

The equivalence of two circuits can, ignoring a global phase, be expressed as

$$U_{C'} = e^{i\phi} U_C$$

where U_C is the unitary transformation of the original circuit and is the corresponding transformation of the optimised circuit and Φ is a global phase. In simple terms, we want to **change the way of execution without changing the calculation that the circuit performs**. Optimisation can therefore examine more than one factor.

This also creates an interesting question. Instead of treating a circuit only as a linear sequence of gates, it can be examined as a **network of interactions** in which certain regions are used more, accumulate more dependencies, or require more routing operations. Such a representation can offer additional information on where the greatest cost appears before an intervention is made in the circuit.

This observation constitutes an important point of transition from the fundamental principles of quantum computing to the problem that will be examined experimentally in the remainder of the work.

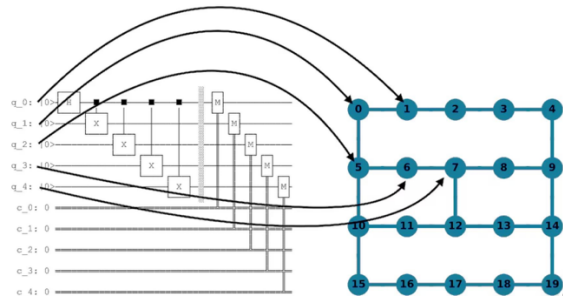


Figure 3.7 Example of quantum circuit mapping and routing according to the connectivity constraints of the physical device.

3.9 Quantum Parametric Circuits and VQE

We define Quantum parametric circuits as the category of quantum circuits whose gate operation is determined by the action of parameters θ . The quantum state is represented by the mathematical formalism:

$$|\psi(\theta)\rangle = U(\theta) |0\rangle^{\otimes n}$$

Where it shows the quantum state produced by the parameter transformation θ , on the initial state of a Qubit. To define an example of these circuits, we will use a quantum-inspired concept of the Variational Quantum Eigensolver. The reason for the existence of this concept lies in the fact of classical optimization in quantum types of circuits with the expectation of minimizing the expected value of a Hamiltonian vector H (Peruzzo et al., 2014).

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$$

In practical implementations, the repeatability of parametric layer executions is used so that their structure remains compatible with the constraints of the available hardware (Kandala et al., 2017). In this paper, in a later chapter, the contribution of repeatability is considered particularly important for pattern recognition as a proposed memory mechanism which we will examine in more detail below.

3.10 Structured and Random Quantum Circuits

Unlike parametric circuits which exhibit organised structural regularity, random quantum circuits define a random generation process

that selects the gates and interactions within the circuit, limiting the reappearance of stable patterns (Bouland et al., 2019).

In this paper, it is appropriate to make this distinction as it will be one of the types of circuits for the experimental verification of the optimisation results that we will conduct in a later chapter.

4. Proposed Adaptive Quantum Optimisation Architecture

4.1 Overview of the Adaptive Architecture

The proposed **Adaptive Quantum Optimisation Architecture** is a hybrid framework for the design and dynamic compilation of quantum circuits. Its goal is to minimise the depth and number of two-qubit gates (such as CNOT) in NISQ devices without compromising the computational accuracy of the algorithm (Preskill, 2018).

The architecture is interposed between the classical optimiser and the quantum processor, acting as a "cognitive" gate reorganisation mechanism. Instead of static execution of a predefined quantum circuit, the system analyses the circuit as a dynamic flow of operators, rearranges the gates based on algebraic commutativity properties, and cancels redundant gate sequences in real time (Nam et al., 2018).

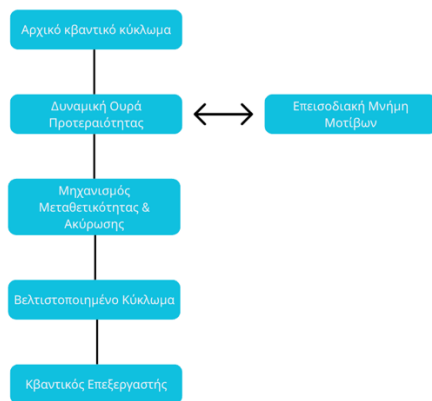


Figure 4.1: Diagrammatic representation of the optimisation system

4.2 Mathematical Model & Cognitive State

The mathematical formalism of the quantum system with n Qubits is described in the Hilbert space of dimension 2^n . A quantum circuit is expressed as the sequence of m unitary transformations $g_i \in SU(2^n)$:

$$U_{\text{total}} = \prod_{i=1}^m g_i = g_m \cdot g_{m-1} \cdots g_2 \cdot g_1$$

In the above representation of the product of transformations, the parameter U_{total} represents the total Unitary operator that describes the complete transformation of the quantum system, the index i the execution order of the individual gate, and m the total number of quantum gates within the circuit. The parameter the i -th quantum gate as an element of the special unitary group of matrices of dimension $2^n \times 2^n$ with a determinant equal to unity, where the parameter n represents the total number of available quantum bits of the system.

We define as the cognitive state of the optimisation mechanism at a discrete time step t the quadruple of variables:

$$S_t = (Q_t, \mathcal{M}_t, \theta_t, \mathcal{F}_t)$$

In this mathematical structure, the parameter S_t expresses the total cognitive state of the system at time t . The variable t is the discrete index of the time step of the optimisation process. The parameter represents the Q_t dynamic priority queue of quantum gates to be processed and executed at time t . The parameter symbolises the episodic memory, which stores the historical simplification patterns, reduction rules, and circuit shortcuts. The vector is the set of k continuous parameters or $\theta_t = (\theta_1, \theta_2, \dots, \theta_k)$ rotation angles for the variable quantum gates at time t , where k is the total number of parametric angles. Finally, the parameter $\mathcal{F}_t$ expresses the estimated value of the quantum fidelity of the state.

The evolution of the cognitive state is governed by the transition operator $\mathcal{T}: S_t \rightarrow S_{t+1}$, as applies algebraic transformation rules to the queue with the aim of maximising fidelity . $\mathcal{F}_t$

The cognitive dimension of the proposed architecture is not intended to be an absolute identification with human cognition, but refers to the technical implementation of selected cognitive properties that approach concepts such as episodic memory and internal state awareness for the purposes of adaptive optimisation. Consequently, the process involves the recognition and storage of the internal state, its recall in case of mathematical identity, and finally the adaptation of the cognitive agent's behaviour based on previous experience. Such properties enjoy the experimental confirmation and fidelity required for dynamism and the reduction of circuit depth and gates in certain cases and are key elements of cognitive architectures (Anderson, 2007; Kotseruba & Tsotsos, 2020).

4.3 Dynamic Priority Queue

The practice of the priority queue with a dynamic processing order of the gates before assignment to the quantum processor, where for each gate of the quantum circuit, the dynamic priority value $P(\cdot)$ is calculated via the function:

$$P(g_i) = w_1 \cdot D(g_i) + w_2 \cdot C(g_i, g_{i+1}) + w_3 \cdot (1 - \epsilon(q_{\text{target}}))$$

In this calculation, the parameter $P(g_i)$ represents the priority value for the gate g_i . The parameter g_i expresses the quantum gate under consideration in the queue, while the parameter g_{i+1} symbolises the immediately following adjacent gate in the queue sequence. The parameter $D(g_i) \in \mathbb{N}$ expresses the integer topological depth of the gate g_i in the graph of circuit dependencies. The parameter is the binary commutativity index, which takes the value 1 if the gate commutes algebraically with its neighbour g_{i+1} and the value 0 otherwise. The parameter $\epsilon(q_{\text{target}}) \in [0, 1]$ captures the nominal error rate of the physical quantum bit or target bits on which the gate acts, as derived from the processor calibration data.

The parameter q_{target} determines the specific quantum bit or pair of bits on the physical hardware. The coefficients $w_1, w_2, w_3 \in [0, 1]$ are positive weighting factors of the individual terms, for which it strictly holds that their sum equals unity, i.e. $w_1 + w_2 + w_3 = 1$. Gates with a higher value are promoted to the beginning of the queue for immediate interaction and potential cancellation with neighbouring operators.

However, while the priority value determines the order in which the circuit gates are examined, it does not in itself constitute the complete mathematical representation of the cost function but is a marking operator regarding the sequential order that will be optimised subsequently based on the general purpose of the process represented as:

$$\min_{C'} J(C') = \lambda_G G(C') + \lambda_D D(C') + \lambda_2 G_2(C')$$

Which is subject to the constraint of maintaining physical circuit behaviour before and after the application of unitary operators:

$$\boxed{U(C') = e^{i\phi} U(C)} \quad \text{where } \phi \in \mathbb{R}$$

While the parameter $J(C')$ expresses the total computational cost of the optimised circuit, the total number of gates, the depth of the active circuit ($D(C')$), the number of successive execution levels, and the number of gates acting on 2 qubits, which are of particular

interest due to their increased cost and sensitivity to errors in modern quantum systems.

The parameters λ are non-negative weighting coefficients that determine the importance of each individual criterion. A larger value of a coefficient increases accordingly the contribution of the specific feature to the total cost.

4.4 Commutativity & Cancellation Mechanism

The core of the architecture relies strictly on Pauli algebra, on the properties of quantum gates, as analysed by Barenco and colleagues in 1995.

The first basic pillar is the Commutativity rule which is applied to the gates , according to the commutator ,

$$g_i \ g_j \quad [g_i, g_j]$$

$$[g_i, g_j] = g_i g_j - g_j g_i = 0 \Rightarrow g_i g_j = g_j g_i$$

In the above-mentioned equation, the commutator denotes the sequential application of the two gates , where according to the difference of the products when it equals the

zero operator, then the given gates commute and their execution order is swapped without burdening the final computational result. For gates acting on the same quantum bits, Pauli algebraic relations are used.

The Second pillar refers to the cancellation and merging rule. For self-inverse gates, such as those of Pauli-X, Pauli-Y, Pauli-Z, Hadamard and CNOT, provided that the condition holds in the circuit where the identity operator, which leads to complete cancellation when two identical gates are found sequentially. Regarding parametric rotations around the Bloch sphere, the phase merging property applies:

$$R_n(\alpha) \cdot R_n(\beta) = R_n(\alpha + \beta)$$

The rotation operator is represented as:

$$R_n(\theta) = e^{-i\theta(\mathbf{n} \cdot \boldsymbol{\sigma})/2},$$

Where the unit vector that determines the rotation axis (e.g. X, Y, Z) and θ the angle in radians, and $\boldsymbol{\sigma}$ the Pauli vector where $\sigma = (X, Y, Z)$ and naturally for the angles α, β it holds that $\alpha + \beta = 2k\pi, k \in \mathbb{Z}$.

However, for angles α, β in the case of a global phase change, i.e. for odd values of k, the result differs by a phase of -1; this phase does not affect the probability distributions of the measurement results and therefore they can be removed by equivalence.

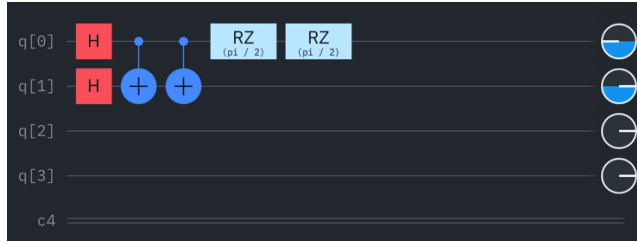


Figure 4.2: Initial circuit as an example of Cancellation and Merging using IBM's Quantum Composer.

4.5 Episodic Memory & Interpretability

The presence of the cognitive property of episodic memory of the term M_e which delimits the **experience** of the agent within the framework of the structural determination of its optimisation effect, is defined as:

$$M_e[k] = r$$

Where **k** is the subtractive key for matching local patterns which consists of the absolute numbers of Qubits which are normalised according to their relative position within the examined pattern and **r** the identifier of the transformation rule of the associated pattern. Additionally, for each examined gate, information is stored regarding whether it possesses a parameter, without the need to store the specific numerical value, so that structurally equivalent patterns can be mapped to the same key even if they appear in different parts of the circuit.

Recall in memory with $O(1)$ complexity is a successful match recorded as **H**, while an incorrect one as **M**, so we can define the successful recall ratio:

The interpretability of the process arises from the direct mapping between the recognised structural pattern and the transformation rule

$$R_{hit} = \frac{H}{H + M},$$

applied. For each relevant decision, the pattern that was recognised, the rule that was activated, and whether the match came from a previous memory entry can be recorded. This process allows for the retrospective examination of transformations without requiring the interpretation of an opaque statistical model.

4.6 Maintaining Functional Equivalence

For the Maintenance of Functional Equivalence between the final and initial circuit, the cognitive agent applies the optimisation rules provided they maintain the condition of the functional behaviour of the original quantum circuit. Let C be the initial circuit and C' the produced circuit, the constraint we mentioned in the previous chapter is written as follows:

$$U(C') = e^{i\phi} U(C),$$

And it maintains exactly this normalisation and sets the optimisation under conditions into effect, as we also see in the corresponding schematic via IBM's Quantum Composer in Figure 4.3. While in addition the properties of phase embedding and symmetric removal that we mentioned also apply, the effect of optimisation under noise is examined for this reason experimentally via Hellinger fidelity in Chapter 5.

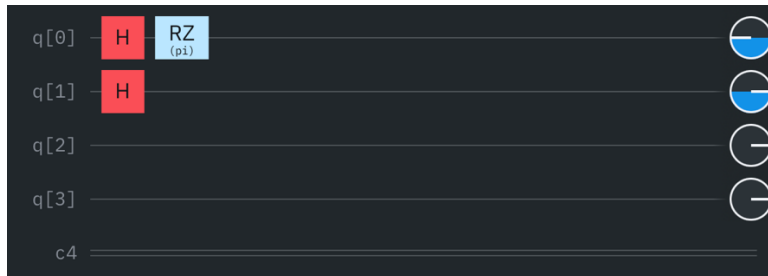


Figure 4.3: Optimised circuit with the proposed optimisation architecture using IBM's Quantum Composer

5. Experimental Evaluation of the Proposed Method

5.1 Purpose, Research Questions and Hypotheses

The main ambition of the experimental execution is to examine to what extent the integration of episodic memory as a cognitively inspired function, in a quantum circuit optimisation process, can utilise cognitive experience from matching previous patterns and maintain the functional coherence of the circuit.

Therefore, the central question of this paper is formulated as follows

Can a cognitively inspired Episodic Memory mechanism utilise optimal swaps and gate additions/removals in a circuit based on repeating structural patterns and if so, is there a benefit to this

differing between structured and corresponding random circuit cases?

For the analysis of the above question, **three testable hypotheses** are given,

The **first Hypothesis** predicts that in corresponding parametric circuits, the proposed H_1 , H_2 , H_3 architecture achieves a greater reduction in the number of applied transformations and depth compared to the IBM Quantum L3 industry standard of Qiskit. The **second Hypothesis** rests on the fact that the repeating structure of parametric circuits will lead to an increased reappearance of already recorded patterns in episodic memory. The **third Hypothesis** represents the potential advantage of the method and its limitation in circuits with less structural repeatability such as random circuits.

5.2 Experimental Framework, Reference Circuits and Metrics

The experimental execution is carried out using Qiskit and the **Qiskit Aer Backend**, while for the execution of the optimisation in **non-ideal** noise conditions based on quantum processor hardware, **FakeEagleV2** is used via the AerSimulator.

```
----- EXPERIMENTAL ENVIRONMENT SETUP -----
Qiskit Version      : 2.5.2
Qiskit Aer Version  : 0.17.2
Target Backend      : fake_eagle
Physical Qubits     : 127
Basis Gates         : ['x', 'sx', 'reset', 'rz', 'measure']...
```

Figure 5.1: Initialisation of the IBM FakeEagleV2 noise simulation experimental infrastructure in the terminal.

In the experimental framework of the paper, optimisation was performed on two different categories of circuits. The first consists of the expected positive result of the method, i.e. it includes a VQE Ansatz 12Q quantum parametric circuit based on EfficientSU2, with eight repeating layers, therefore we examine Hypothesis 2 and the relative advantage of the architecture against its repetitive structures.

The second category includes the structure of a random unstructured circuit and examines Hypothesis 3 and its limitation against the non-linearity of the random circuit and how episodic memory fails in relation to the industry standard.

Naturally, for the first Hypothesis to hold, the set operation must also be satisfied
i.e. the union of the two hypotheses () must verify the .

$$H_1 = H_2 \cup H_3 \quad H_2, H_3 \quad H_1$$

Before applying individual rules of the proposed optimiser, we define the base set of gates allowed in the circuit as follows:

$$B = \{CX, RZ, SX, X\},$$

While the evaluation consists of 6 main parts, the total gate count, the circuit depth, the count of Two-Qubit gates, the optimisation time in sec and the Hellinger fidelity where the result distribution probabilities are calculated in 2000 executions of the corresponding circuit, with N=30 control execution times for each condition and seeds from 42 to 71.

This therefore leads us to the mathematical cost function model that we also mentioned in Chapter 4.

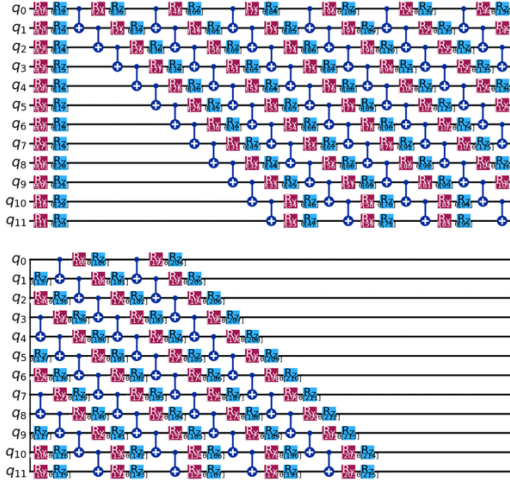


Figure 5.2: Visual representation of the structure of the parametric circuit VQE_Ansatz_12q (EfficientSU2, 12 qubits) before the compilation process.

5.3 Implementation and Operation of Cognitive Memory

The approach of Episodic memory and the transformations included in the architecture allows for the recall of repetitive structures and is therefore linked to the use of previous experience in cognitive architectures (Anderson et al., 2004; Kotseruba & Tsotsos, 2020).

Following experimental execution, the only stored pattern that applied its individual transformation is the rotation phase embedding rule.

In the VQE circuit, 107 memory recalls and one entry were recorded, with while in the case of the random circuit execution, 33

Recalls and one entry were recorded, with . Therefore, in the structured VQE, $R_{hit} = 99,07\%$

a total of 108 occurrences of the pattern were produced against 34 in the random circuit. The greater repetition is compatible with the layered structure of parametric circuits used in VQE and other hybrid quantum algorithms (Kandala et al., 2017).

Finally, the use of local algebraic unitary transformations for the reduction of a quantum circuit enjoys full confirmation regarding peephole techniques

optimization. Liu et al. (2021), for example, reported up to 18% reduction in CNOT gates and an average of 11.7% in circuit optimisation experiments, showing that local rewrites can have a measurable effect on the structure of the final circuit. This philosophy is also compatible with the operation of the Qiskit transpiler, where optimisation is based on rewrites, synthesis, and iterative passes over the circuit.

In the present implementation, however, the high rate of occurrences is not necessarily interpreted as computational acceleration, but as an indication of the reappearance of an already known structural pattern. The effect of memory on execution time is examined independently in the deconstruction experiment of the next section.

```

--- Πείραμα Κλιμάκωσης Layers VQE (Memory Scaling Analysis) ---
Layers: 1 | Total Gates: 131 | Hits: 23 | Misses: 238 | Hit Ratio: 8.81% | Time: 0.0015s
Layers: 2 | Total Gates: 202 | Hits: 35 | Misses: 368 | Hit Ratio: 8.68% | Time: 0.0019s
Layers: 4 | Total Gates: 344 | Hits: 59 | Misses: 628 | Hit Ratio: 8.59% | Time: 0.0035s
Layers: 8 | Total Gates: 628 | Hits: 107 | Misses: 1148 | Hit Ratio: 8.53% | Time: 0.0047s

```

Figure 5.2: Results of the layer scaling experiment execution in the terminal.

5.4 Experimental Scenarios

The Architecture of the cognitive optimiser showed different behaviour across the range of different circuits in which the mechanism of the proposed optimisation was introduced, with the two experimental scenarios being a characteristic example. In VQE, it produced 520 gates and a depth of 61, against 628 gates and a depth of 70 for Qiskit's optimisation level 3. The successful reduction corresponds to 17.20% in the gate count and 12.86% in the circuit depth, while the CX gate count remained equal at 88, so we had a 0% change due to the optimiser's policy. Contrary to the expected results, in the random circuit, the gate reduction was limited to 3.33%, the depth increased from 110 to 120, and the CX gates from 96 to 112.

The gap in the agent's behaviour is also compatible with similar quantum circuit optimisation studies, during which the effectiveness of optimised transformations depends on the available circuit structure and the choice of local optimisation method. Liu et al. (2021) reported up to 18% reduction in CNOT gates via relaxed peephole optimisation, while techniques such as Quartz utilise equivalent local rewrites to search for lower-cost circuits (Xu et al., 2022). The results of this work are not directly comparable numerically due to different circuits and cost functions, but they support the importance of exploiting internal structure during quantum circuit optimisation.

C i r c u i t	M e t h o d	G a t e s	D e p t h	C X	
VQE	Qiskit L3	628	70	88	
VQE	P r o p o s e d	5 2 0	6 1	8 8	

C i r c u i t	M e t h o d	G a t e s	D e p t h	C X
Random	Qiskit L3	481	110	96
R a n d o m	P r o p o s e d	4 6 5	1 2 0	1 1 2

Table 5.1: Comparison of structural metrics

Regarding Hellinger fidelity, the 30 consecutive iterations on each parameter confirm in VQE $F_H = 0.7937 \pm 0.0080$, while in the random circuit, it resulted in 0.4894 ± 0.0194 for the previous method and 0.4887 ± 0.0145 for Qiskit L3. Consequently, the VQE reduction was not accompanied by an observed loss of fidelity in the specific simulation protocol. Evaluation under noise is particularly important for NISQ circuits, as depth and quantum gates are key limiting factors of execution (Preskill, 2018; Kandala et al., 2017).

5.4.1 Memory deconstruction experiment

The overall experimental repeatability of the episodic memory was 500 pairs of Memory ON/OFF executions, with timing of the optimisation process. Disabling the memory did not change the structural metrics, confirming that the gate and depth reductions originate from the algebraic rules and not from the memory. In VQE, the median time overhead of the memory was 0.097 ms, while in the random circuit, it was 0.033 ms. The corresponding median time increases were 2.47% and 0.75%.

The result shows that the memory in the present implementation functions as a recall and traceability mechanism and not as an acceleration mechanism. This distinction is important, as it experimentally isolates the contribution of the memory from the contribution of the optimisation rules.

Index	VQE Original	VQE Qiskit L3	VQE Cognitive	Random Original	Random Qiskit L3	Random Cognitive
T o t a l g a t e s	3 0 4	6 2 8	5 2 0	1 3 4	4 8 1	4 6 5
D e p t h	4 3	7 0	6 1	2 0	1 1 0	1 2 0
C X g a t e s	8	8	8	8	0	9 6
R_{hit}	—	9 9 . 0	7 %	—	—	9 7 . 0 6 %
F_H , mean value $\pm$ Deviation	—	0.7937 $\pm$ 0.0080	0.7937 $\pm$ 0.0080	—	0.4887 $\pm$ 0.0145	0.4894 $\pm$ 0.0194
Gate change vs L3	—	—	-17.20%	—	—	-3.33%

Index	VQE Original	VQE Qiskit L3	VQE Cognitive	Random Original	Random Qiskit L3	Random Cognitive
Depth change vs L3	–	–	2.86%	–	–	+9.09%

Table 5.2: Comparison of structural metrics, Hellinger fidelity, and episodic memory behaviour for VQE and Random circuits.

5.4.2 Statistical evaluation of the memory effect

To isolate the temporal effect of episodic memory, 500 consecutive executions were performed for both circuits, with sequential enabling and disabling of their cognitive function. Timing was restricted exclusively to the optimisation process. In VQE, the median time difference was 0.097 ms, which

$$T_{on} - T_{off}$$

corresponds to an overhead of 2.47%, while in the random circuit it was limited to 0.033 ms and 0.75%, respectively.

The Wilcoxon signed-rank test was applied to the paired time measurements, as the analysis is based on the differences between the two conditions and extreme time values occur. The difference was statistically significant for both VQE ($p < 0.001$) and the random circuit ($p < 0.001$). At the same time, memory activation did not alter the gate count, depth, or CX gates. Consequently, in the present implementation, episodic memory provides a recall function for repeating patterns with a small but measurable time cost, rather than computational acceleration.

Circuit	Median ON	Median OFF	Median ΔT	Change	Wilcoxon p
VQE	4.169 ms	4.068 ms	+0.097 ms	+2.47%	<0.001
Random	5.111 ms	5.073 ms	+0.033 ms	+0.75%	<0.001

Table 5.3: Results of the episodic memory ablation experiment

The median times for VQE and Random are derived from the 500 iterations of the experiment (Wilcoxon, 1945).

5.5 Evaluation of Research Hypotheses

The results are now logically connected to Hypothesis for the H_1 VQE circuit, where the proposed optimisation method produced a smaller circuit than Qiskit L3. The limited improvement and increase in depth in the random circuit, which

is part of the hypothesis H_3 regarding the limitation of the optimiser on random architectures due to the lack of pattern repeatability, was expected from the outset.

Finally, H_2 supported regarding the recognition of repeating patterns as VQE presented more recalls of the stored pattern. However, the ablation experiment showed that memory is not the cause of structural compression nor the acceleration mechanism in the current implementation.

Hypothesis	Result	Conclusion
H_1 — Greater structural compression in VQE versus Qiskit L3	Reduction of gates and depth in VQE	Supported
H_2 — Increased pattern recurrence in the structured circuit	107 recalls in VQE versus 33 in Random	Supported for the examined pattern
H_3 — Limitation of the advantage in a circuit with lower structural repeatability	Small reduction in gates but increase in depth and CX in Random	Consistent with the results

Table 5.4: Summary evaluation of research hypotheses based on experimental results.

6. Discussion and Conclusions

6.1 Interpretation of Experimental Results

The results show that the performance of the proposed method depends on the circuit structure. The clearest compression appeared in the structured VQE, while in the random circuit the benefit in gate count was limited and accompanied by an increase in depth and CX gates. This observation is consistent with optimisation approaches that leverage local equivalences and repeating sub-circuits, the effectiveness of which depends on available rewriting opportunities (Liu et al., 2021; Xu et al., 2022).

The maintenance of corresponding Hellinger fidelity in VQE shows that the structural reduction was not accompanied by observed degradation in the examined noise model. This result is of particular importance for NISQ circuits, where depth and quantum operations are key constraints of practical execution (Preskill, 2018; Kandala et al., 2017).

The ablation experiment also clarified the role of episodic memory. The memory recognised the recurrence of the stored pattern, without, however,

being the cause of the structural compression. Its activation incurred a small but statistically significant time cost. Therefore, in the current architecture, its main contribution lies in the recall and traceability of previous patterns and not in the direct acceleration of optimisation.

6.2 Cognitive Dimension of the Proposed Architecture

The dimension of episodic memory in a cognitive dimension is located in the maintenance of internal state and recall of patterns from its short-term memory. This approach follows the functional understanding of cognitive architectures, where memory, state, and the utilisation of previous experience constitute distinct processing mechanisms (Anderson et al., 2004; Kotseruba & Tsotsos, 2020).

Experimental verification proved that the memory recognises repeating structures without causing circuit compression or computational acceleration. In conclusion, the proposed optimisation method is characterised as a **cognitively inspired optimisation architecture** and not as a complete model of artificial cognitive function. This distinction defines the contribution of the work and avoids the attribution of cognitive properties that were not experimentally substantiated.

6.3 Scientific and Technical Contribution

The main contribution of the work is the development and experimental execution and evaluation of a cognitively inspired agent that combines algebraic transformations with abstract episodic memory. Unlike purely static rewriting processes, the architecture maintains a history of recognised patterns and allows their recall during optimisation.

At the same time, the work experimentally separates the contribution of optimisation rules from that of memory through an ablation experiment. This result offers a clearer picture of both the capabilities and the limits of the proposed cognitively inspired approach.

6.4 Limitations and Future Extension

The present evaluation is limited to two experimental scenarios and a simulated noise model; therefore, the results cannot be generalised to all categories of quantum circuits or to real quantum hardware. Furthermore, the episodic memory recalls already recognised rules without autonomously learning new transformations.

As a future extension of the research, evaluation on more architectures and real quantum devices constitutes a scientifically interesting option. Of particular

interest is the development of memory that would dynamically influence the selection and priority of transformations, turning previous experience into an active factor of optimisation.

6.5 Final Conclusions

The work investigated whether cognitively inspired mechanisms can be integrated into quantum circuit optimisation. The results showed that the proposed method can leverage repeating structure, with the most significant benefit appearing in the examined VQE circuit.

At the same time, the episodic memory functioned as a mechanism of recall and traceability, without being the cause of the structural compression. Consequently, the work does not propose a complete model of artificial intelligence, but a first computational framework for the study of cognitively inspired mechanisms in the field of quantum computing.

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